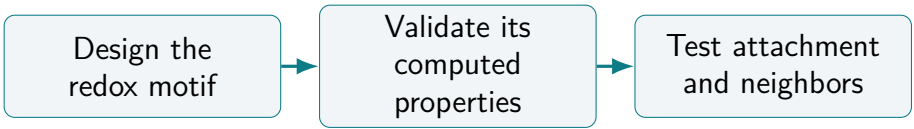
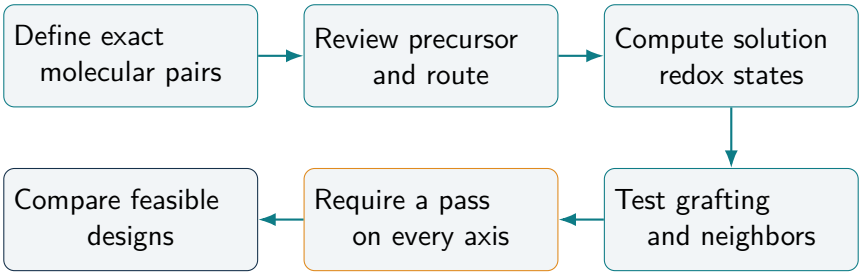
Which redox groups remain useful after grafting onto a polymer?



Candidate expansion and validation: progress through 26 October 2026

The target is a useful grafted material, so molecular advantages must survive attachment and polymer context.

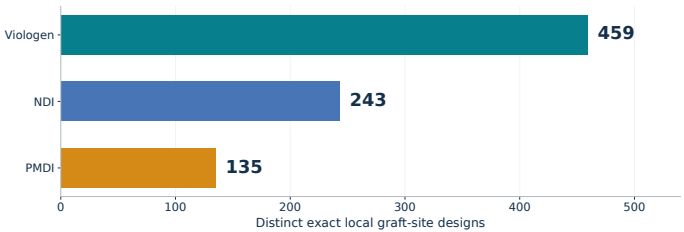
One exact design links the whole workflow



Exact identity and separate validation gates connect expansion to selection.

Current route: post-functionalization of a preformed benzyl-chloride polymer. Explicit method controls are kept separate.

The library is broad within three reduction-active cores



837
exact local designs

1,593
generation recipes

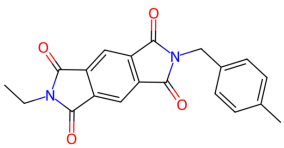
32
overlap designs with
both computed potentials

Enumeration is systematic; equivalent recipes are retained while each exact product is counted once.

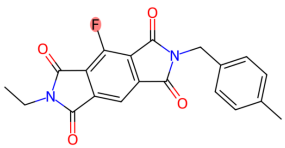
33 library graphs overlap the 75-candidate evaluated cohort; 32 have both computed potentials. This is not an 837-molecule quantum screen.

A matched comparison asks one chemical question

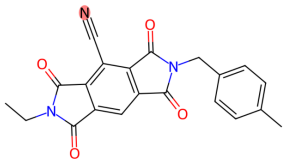
Baseline PMDI



Ring F



Ring CN

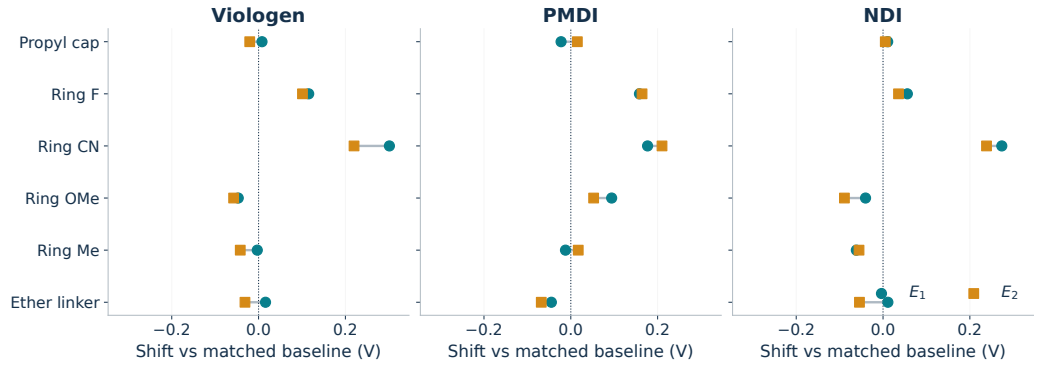


Same PMDI core, ethyl cap and local benzyl attachment; declared ring substitution changes.

Attach every variation to an exact baseline before interpreting its computed effect.

Exact graft-site molecular graphs from the checked candidate feed. The corresponding precursor graphs are retained separately.

Ring electronics shifts voltage more than these cap changes



18 systematic contrasts quantify tuning across all three cores; they do not establish better material performance.

Matched solution potentials, each relative to its declared ethyl baseline. Circles: first step; squares: second step. All 26 cohort contrasts reproduce.

The next expansion should fill chemistry gaps

Seven core families

Viologen PMDI NDI

BQ AQ

Phenothiazine TEMPO

Current systematic coverage emphasizes the first three.

Predeclared polymer-design comparison

	Flexible	Constrained
Baseline	☐	☐
Shielded	☐	☐

Four slots per core: 28.
Six extra electronic contrasts: 34 total.

Test shielding and linker constraints as tradeoffs: reduced association can cost attachment access or capacity.

34 planned slots, not 34 validated candidates. Partial PMDI, anthraquinone and phenothiazine panels are mapped but route-held and uncomputed.

Validation has three distinct jobs

1. Numerical correctness

Is the stated quantity computed consistently?

Geometry, spin, Hessian and balanced-cycle checks.

2. Experimental accuracy

Does it reproduce measured properties?

39 redox endpoints for exact free molecules in MeCN.

3. Model robustness

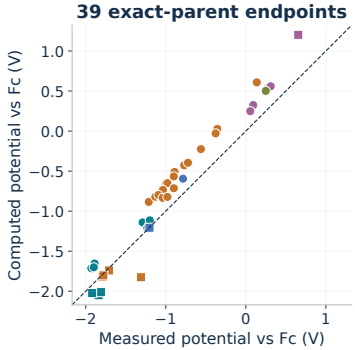
Does the conclusion survive reasonable choices?

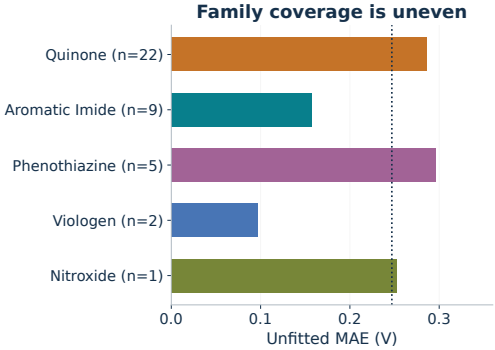
Basis, rotor treatment, reference, conformers and speciation.

A calculation can be numerically correct while its chemical prediction remains uncertain.

The candidate reconstruction reports 628 arithmetic checks with zero discrepancies. These are bookkeeping checks, not experimental validation.

The measured redox benchmark is complete: 39 / 39



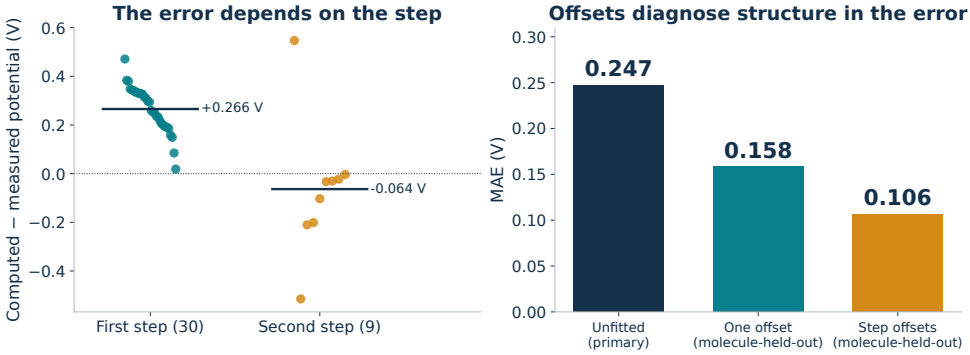


Unfitted MAE 0.247 V; RMSE 0.279 V. 30 molecules, 7 studies, MeCN.

This establishes a free-molecule error scale; exact grafts and polymer interactions need separate validation.

Retrospective panel; molecular-cluster 95% MAE interval: 0.213–0.286 V. No NDI endpoint; a separate NDI challenge is sourced but uncomputed.

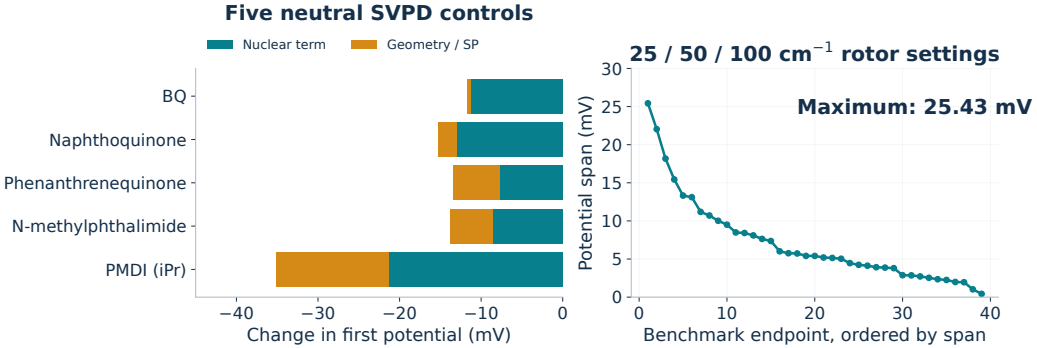
The benchmark error is structured, not just random



The first/second-step difference motivates reference and charge-state diagnostics; one Fc shift cannot fix step spacings.

Offset results are post hoc, molecule-held-out internal diagnostics. Primary potentials remain unfitted; a causal Fc correction is not established.

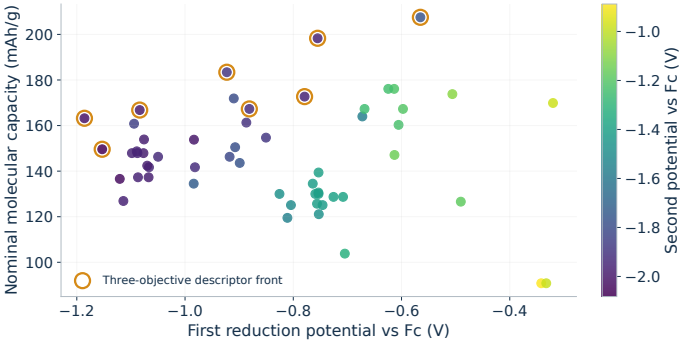
Basis and thermal controls quantify specific sensitivities



These tested settings move potentials by tens of mV; missing conformers and experimental accuracy remain separate questions.

Neutral SVPD reoptimization/Hessians with existing anion and Fc fixed. Rotor spans use fixed structures, energies and spins. These are not prediction intervals.

A descriptor leader still has to pass every required axis



8

descriptor-front models

0

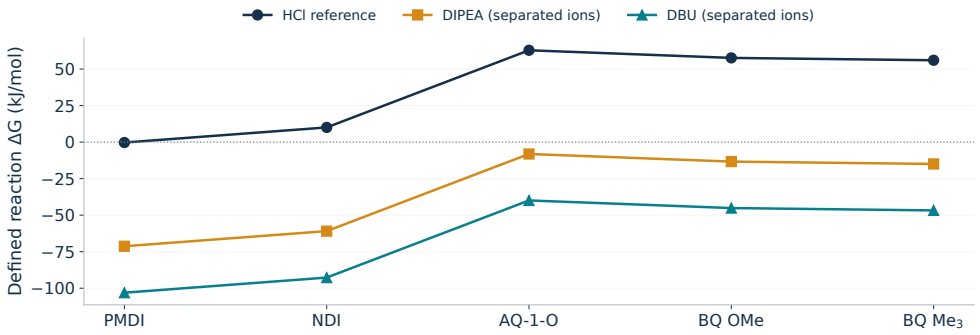
all-axis eligible

Route, stability, accessibility and neighbor evidence remain incomplete.

A strong voltage or nominal capacity cannot compensate for an unacceptable attachment route or unwanted interaction.

Projection of a three-objective comparison over 56 molecular models. Nominal molecular capacity is not usable polymer capacity. Four application cutoffs remain unset.

A defined base shifts grafting thermodynamics across the panel

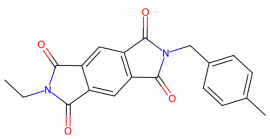


For a fixed base/byproduct model: $\Delta G_{\text{base}} = \Delta G_{\text{HCl reference}} + C_{\text{base}}$.

Keep base-required BQ candidates for testing; ranking is unchanged and absolute reaction signs remain conditional.

DIPEA UV: log $K^{\text{H}} = 18.64(1)$ in MeCN at 25 °C; DIPEAH/PF_6^- NMR: pKa 18.7 in MeCN-d_3. Conditions differ; no yield or barrier prediction.

One PMDI design now connects the molecular and polymer tests



Local model: 4-methylbenzyl/ethyl PMDI.

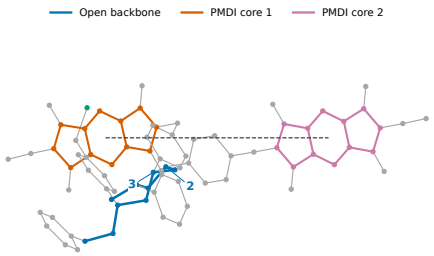
Redox in MeCN	$E_1 = -1.076$, $E_2 = -1.977$ V
Precursor access	Exact source identified
First graft, DBU	-97.72 / -96.96 kJ/mol
Neighbor cycle	-1.86 kJ/mol

Selectivity and redox retention after attachment remain unresolved.

The framework can trace one design through the calculations while retaining a conditional route decision.

First-graft values: sites 2 / 3 of a five-repeat model, separated ions. One minimum per graph; absolute signs retain base/speciation limits.

The polymer cycle measures a conditional neighbor effect



3D core-centroid distance: 16.30 Å; plane angle: 69.0°

$$\Delta\Delta G = -1.86 \text{ kJ/mol}$$

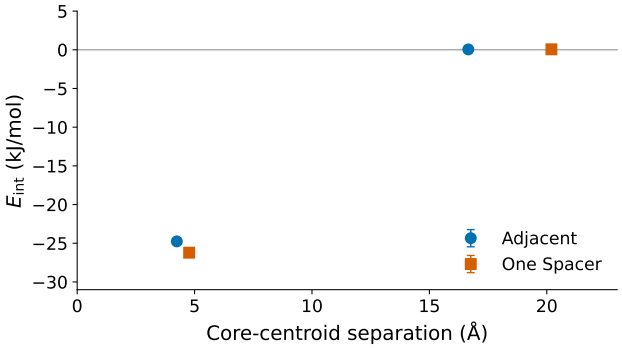
Core separation: 16.30 Å.

Finite PMDI cycle in MeCN; one minimum for each graph.

This is whole-chain nonadditivity, not isolated motif binding or electronic coupling. Conformer convergence is still open.

$\Delta\Delta G = G_{11} + G_{00} - G_{10} - G_{01}$. Native Hessians and all four state sums independently reconstructed.

Close motifs can interact even with a spacer



Close poses

-24.76 / -26.22 kJ/mol

Separated poses

approximately zero

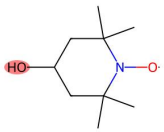
Adjacent and one-spacer phthalimide controls.

$E_{\text{int}} = E_{AB} - E_A - E_B.$

A spacer can still permit an attractive close contact; weak unwanted intersite interaction remains a design goal.

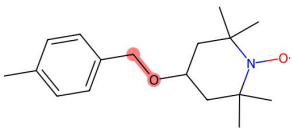
Four frozen capped-fragment poses; MeCN electronic energies. Cap-length ranges are not uncertainty intervals. No basis-superposition correction, populations, binding free energy or electronic coupling is established.

TEMPO: a route can pass while the material decision waits



4-Hydroxy-TEMPO precursor

Precursor



4-Methylbenzyl local graft model

local graft model

4-O attachment on a preformed Merrifield-type support.

Route screen
Proceed

Local MeCN model
$+0.579$ V vs Fc/Fc^{+}

Material decision
Pending
9 required axes unresolved.

A documented attachment and a computed potential are useful, but neither establishes all-axis feasibility.

US6441243B1, Example A4: 4-hydroxy-TEMPO on Merrifield resin. Local model is not the measured bead; grafting ΔG and interactions remain uncomputed.

Next: convert molecular leaders into defensible material choices

Define useful performance

Set the operating window and minimum material capacity.

Declare tolerable unwanted association and intersite coupling.

Broaden the chemistry

Fill the seven-core matched comparison panel.

Review exact routes before expensive redox campaigns.

Bound polymer uncertainty

Finish conformer and backbone-configuration comparisons.

Distinguish interaction energy, grafting nonadditivity and coupling.

We have a linked, auditable workflow and concrete results; a synthesis shortlist requires all-axis evidence.

Backup: what the calculations actually contain

Stage	Stated model
Warm start	Gas-phase UMA; execution mode, model version, charge and spin retained
Geometry	r^2SCAN-D4 / def2-SVP for neutral/cation, def2-SVPD for anions; SMD MeCN
Electronic energy	Uniform ωB97M-V / def2-TZVPD, VV10, MeCN
Nuclear term	Native DFT Hessian; projected internal modes; qRRHO at 298.15 K
Reference	Matched Fc/Fc^+ model and stated standard states

Gas-phase ML screening is an upstream accelerator. It does not replace the reported DFT/SMD free energies.

Backup: QC gates have a precise, limited scope

Geometry / identity	Actual graph, graft bond, open-chain connectivity, stereochemistry, stationary structure
Electronic state	Explicit charge/multiplicity; converged SCF; $\lvert\langle S^2\rangle - S(S+1)\rvert \leq 0.1$
Hessian	Finite and symmetric; sampled finite-difference test: two columns at 0.0005 / 0.001 bohr, 1% relative tolerance
Internal modes	Modes below -5 cm^{-1} fail; tolerated $[-5, 0)$ modes enter as $\lvert\nu\rvert$; frequencies unscaled
Partition	Projected native modes; xTB used as qRRHO integrator and symmetry detector, not as the Hessian source
Force	Gate 4.5×10^{-4} Ha/bohr; actual checked PMDI pair: 7.62×10^{-6}

Sampling two Hessian columns is a diagnostic; it does not verify every element of the full matrix.

Backup: external coverage limits interpretation

Family	Endpoints	Molecules	Studies
Quinones	22	18	3
Aromatic imides	9	6	1
Phenothiazines	5	4	1
Viologen	2	1	1
Nitroxide	1	1	1

NDI has no endpoint. The panel is retrospective and shares studies within families.

Do not transfer the overall MAE to every new scaffold, exact graft or polymer quantity as a prediction interval.

Molecule-cluster resampling: 2,000 replicates. Seven unequal study clusters give weakly calibrated study-bootstrap intervals.

Backup: which properties have external validation?

Property	Checked evidence	Remaining limit
Redox potential	39 exact-parent MeCN measurements	Exact graft / NDI coverage
Nominal capacity	Exact formula, declared electron count, arithmetic	Accessible polymer electrons / loading
Charge transfer	Scoped calculated reorganization diagnostics	No validated material rate or transport
Grafting ΔG	Balanced cycles, base pKa / relative controls	Speciation, barriers, selectivity and yield
Neighbor effects	Native Hessians and finite occupancy cycles	Conformer convergence, isolated binding and coupling

The evidence supports different claims for different properties; computed does not mean experimentally validated.

Backup: Nitro installation and numerical qualification differ

Verified

Separate locked installation.

V100 standard inference: 10 states, 40 geometries.

Six stored legacy energies reproduce within 6.10×10^{-7} eV.

Model-version drift traced to composition averaging.

Still separate

Actual Nitro redox-panel parity and matched-device speedup remain pending.

V100 support uses a separate CUDA12 standard-inference profile.

No DFT/SMD free energy was replaced.

The default entry point can prefer Nitro while its redox-specific numerical/performance qualification remains unfinished.

Frozen validation reports: UMA Nitro 23 October; V100 diagnosis 24 October. Do not label V100 fallback execution as Nitro.

Backup: reproducibility extends beyond recomputing a sum

Primary reviewer package

243 thermochemical states
32 electronic-only records

Relocated arithmetic; 242 molecular Hessian reconstructions.

PMDI neighbor addendum

14 raw states
13 molecular Hessians

Nuclear terms and balanced cycles replay after extraction with original paths blocked.

Code, frozen evidence and raw-state replay are prepared; public data deposition/DOI and final metadata remain open.

Packages overlap and their counts must not be added. Cached replay does not rerun the SCF, optimize geometries or establish conformer convergence.

Backup: the evidence behind the presentation

Enumeration	Exact graph deduplication and recipe counts independently recounted
Matched effects	All 26 pairs reconstructed from unrounded candidate potentials
Benchmark	All 39 residuals and aggregate metrics reconstructed; molecule-held-out offsets independently recomputed
PMDI cycle	Four stored G values independently summed using exact SI conversion
Presentation	Evidence JSON, source hashes, editable LaTeX, plotting code, compiler record and separate visual review

Scientific context: Elhajj & Gozem, JCTC (2024); Grimme, Chem. Eur. J. (2012); Kim et al., Chem. Eur. J. (2022); checked full references in the manuscript.

Every quantitative claim has a stated evidence scope; missing properties remain missing.